Reg. No. 1 8 M S R C H 0 1 1

[18MSCH2H03]

SECOND SEMESTER UNIVERSITY EXAMINATION, MAY - 2019

PHYSICAL CHEMISTRY - II

Time: 03 Hours

Maximum Marks:70

Instructions:
I. Question 1 is compulsory, carries 10 marks.
II. Questions 2 to 7 carry 15 marks each, answer any four.

1. Answer any FIVE. 5×2=10

a. Define Eigen value and Eigen function.

b. What is degeneracy? How does it arise?

c. What are top-down and bottom-up methods?

d. Define and classify emulsions.

e. Show that for a zero-order reaction $t_{½}$ is directly proportional to initial concentration of the reactant and inversely proportional to rate constant.

f. Rate constant of a reaction at 700K and 760K are 0.01 $M^{-1}S^{-1}$ and 0.105 $M^{-1}S^{-1}$, respectively. Find E_a.

2. a. Explain de Broglie's hypothesis with experimental confirmation.

b Define and explain the time independent Schrodinger's equation.

c. Prove, if operators A and B commute then they have common Eigen functions. 5 + 5+5

3. a. Derive the commutation relations of the three components of the angular momentum operator. Show their commutation relations with L^2 operator.

b. Discuss Schrodinger's equation to rigid planar rotator. 10+5

4. a. Explain Frankel and Schottky defects with ion migration illustration.

b. What is sol-gel method? List its advantages. 10+5

5. a. Write a note on Donnan membrane equilibrium with illustrations.

b. Explain steric and electrostatic stabilization. 10+5

6. a. Derive kinetic equation for photochemical reaction between hydrogen and bromine using steady state approximation.

b. i. For a first order reaction, show that time required for 99% completion is twice the time required for the completion of 90% of reaction.

ii. Rate of a reaction becomes 1.414 times when concentration of the reactant is doubled. Find the order of the reaction. 10+5

7. a. Explain Langmuir adsorption isotherm with and without dissociation.

b. Explain phase diagram of three component system with an example. 10+5

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